

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed cross-corpus transfer pairs among Chapman-Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap as the operating interval (BCa, $B = 10,000$, re-run on the full 60-experiment grid). The study was governed by a pre-specified protocol with registered, dated amendments, publicly archived with per-file checksums (see the Data availability statement). The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified empirical direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60

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24 (75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95%
 25 CIs containing zero) is *not* met ($23/60 = 38.3\%$), triggering the registered
 26 “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit
 27 ($+0.0159$ vs. $+0.0083$). A validated 27-cell Shapley decomposition separates
 28 slope, intercept, and prevalence effects; a predictability experiment triggers
 29 the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$); and
 30 a discrimination-aware gate (C5) shows 7/14 pair $\times$ architecture strata pass-
 31 ing both calibration and discrimination layers. Together these results bound
 32 when post-hoc recalibration pays off across institutions, and when it does
 33 not.
 34 *Keywords:* temperature scaling, model calibration, electrocardiogram,
 35 dataset shift, cross-corpus transfer, distribution mismatch

36 **Highlights**

- 37 • Pre-registered 6-pair $\times$ 2-architecture $\times$ 5-seed grid quantifies the OOD
38 calibration benefit of temperature scaling in cross-corpus ECG transfer
39 (51/60 supported).
- 40 • Patient-level paired BCa bootstrap; OOD benefit is about $1.9\times$ the ID
41 benefit (means $+0.0159$ vs. $+0.0083$).
- 42 • The pre-registered ID boundary criterion fails ($38.3\% < 80\%$); the reg-
43 istered branch fires, qualifying the headline claim.
- 44 • A discrimination-aware gate (C5) passes in 7/14 strata: recalibration
45 is moot below the majority-class accuracy bar.
- 46 • Label-free deployment gates fail their pre-registered test; the deploy-
47 ment lookup table is an audit-time decision aid.

48 1. Introduction

49 Modern ECG deep models achieve strong in-distribution (ID) accuracy
50 and calibration, but their reliability under distribution shift remains open
51 [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split;
52 its benefit, a reduction in expected calibration error (ECE), is measured on
53 that same distribution and reverses on a shifted target.

54 **Why this matters for ECG.** PTB-XL [3], Chapman-Shaoxing [4], and
55 CPSC differ in acquisition protocols, lead sets, and label vocabularies. A
56 model calibrated at one institution meets covariate, prior-probability, and
57 label-concept shift at the next [23]. Clinicians and regulators need a decision
58 rule: *given my local data budget and the measurable shift, should I deploy the*
59 *recalibrated model, buy labels, or refuse?*

60 **Temperature scaling and pre-registration.** Temperature scaling
61 (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted
62 under a convex loss, it minimizes NLL on the calibration split, so its cali-
63 bration benefit there is empirically small; on a shifted target the source-fit
64 temperature is mismatched to the target logits and a non-trivial benefit can
65 remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant
66 under confidence-shifting monotone maps, and TS can *increase* ECE where
67 the model is under-confident (nine negative cells in our data). It motivates
68 the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned
69 decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing,
70 protocol v2.1-A1 [24] fixes the primary endpoint, test, family structure, and
71 failure branches before unblinding; revision A1 re-locates the endpoint from
72 **decay** to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with

73 before/after protocol hashes [25, 26]. Multi-seed validation (5 seeds $\times$ 6
 74 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed
 75 results, including nine counter-examples, are reported transparently.

76 **Contributions.** Five independent contributions:

77 **C1 (Primary OOD benefit):** On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$,
 78 TS yields a statistically significant positive OOD benefit in 51/60 seed
 79 experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30,
 80 ResNet1D 24/30), with nine counter-examples transparently disclosed
 81 and mechanistically explained. InceptionTime: three of six directions
 82 achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two
 83 4/5, two 3/5.

84 **C2 (Boundary condition, pre-registered branch triggered):** Across
 85 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI)
 86 have 95% CIs containing zero and 34/60 have CI lower bound > 0
 87 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD
 88 mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI
 89 containing zero) is *not* met, triggering the “ID nonzero boundary”
 90 branch (protocol A1.4.2): the premise that TS has no ID recalibrat-
 91 able miscalibration is re-examined, the decay contrast restored as
 92 exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the
 93 ID benefit but is *not* exclusively OOD-specific. $47/60 = 78.3\%$ of ID
 94 point estimates are positive and $2/60 = 3.3\%$ have $\Delta\text{ECE}_{\text{ID}} < -0.01$
 95 (counter-example 6, -0.0152 ; one marginal case, -0.0128).

96 **C3 (Methodological, Shapley decomposition validation):** A 27-
 97 cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence,
 98 $n = 20,000$) recovers ground truth in the identifiable regime (27/27
 99 at $n = 20,000$; 21/27 at $n = 500$), flags entangled regions without a
 100 threshold, and reports bootstrap CIs on absolute Shapley values.

101 **C4 (Practical, method-selection boundary):** Across 8 methods $\times$ 6
 102 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior
 103 adaptation is fragile when ill-conditioned cells ($w_{ID} > 10$) are included
 104 (InceptionTime 15/30); after exclusion the EM OOD finding attenu-
 105 ates. The mechanism is prior-likelihood mismatch: EM fits a target
 106 prior that amplifies shift-induced miscalibration when the source con-
 107 fusion matrix is ill-conditioned.

108 **C5 (Clinical, discrimination-aware calibration gate):** A two-layer
 109 gate decouples calibration effectiveness (C1: $\Delta ECE_{OOD} > 0$) from
 110 clinical viability (accuracy $\geq$ source-domain baseline). Across 14
 111 pair $\times$ architecture strata, 7/14 pass both layers; 7/14 fail the discrimi-
 112 nation layer, including directions with positive ΔECE_{OOD} but accuracy
 113 below baseline (e.g., PTB-XL $\rightarrow$ Chapman ResNet1D: $\Delta ECE = +0.029$,
 114 acc = 0.657 < 0.713). The gate reframes the 9/60 counter-examples:
 115 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures
 116 (C5 rejects despite C1 passing).

117 Prior ECG calibration studies report the phenomenon [10, 11, 18, 19] but
 118 not an actionable *when does TS pay off* answer with multi-seed robustness
 119 evidence. The novelty is *not* multi-seed validation on ECG (a methodologi-

cal choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

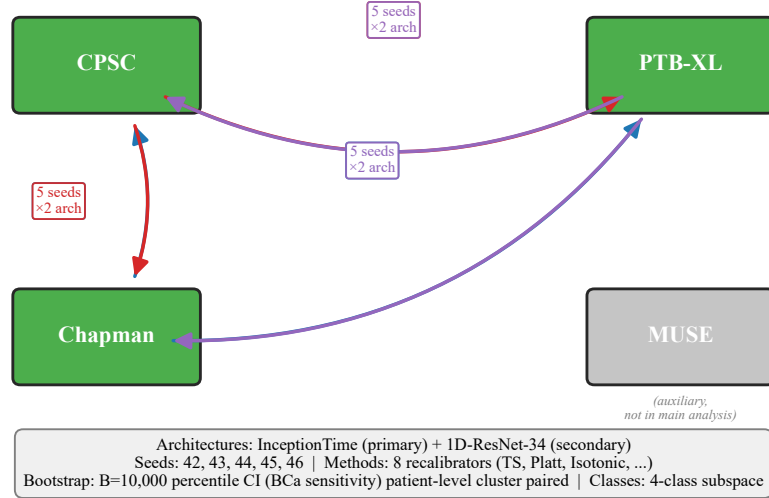


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, 1D-ResNet-34) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

127 2. Related Work

128 2.1. Calibration Estimation and Recalibration

129 The calibration of modern neural networks has been studied extensively
130 since the seminal work of Guo et al. [2], who showed that deep networks
131 are often overconfident and introduced temperature scaling (TS) as a simple
132 post-hoc recalibration method. Nixon et al. [6] proposed the static calibration
133 error, and Kumar et al. [16] introduced verified uncertainty calibration
134 with distribution-free guarantees. Kull et al. [7] generalized TS to Dirichlet
135 calibration for multi-class settings. In the clinical risk-model context, Van
136 Calster et al. [14, 15] established a calibration hierarchy from utopia to em-
137 pirical data and argued that calibration is the “Achilles heel” of predictive
138 analytics. Strodthoff et al. [17] benchmarked deep learning for ECG analy-
139 sis on PTB-XL and noted calibration degradation under subpopulation shift,
140 motivating our cross-corpus design.

141 2.2. Prior-Shift Estimation and Transfer Calibration

142 Lipton et al. [9] introduced black-box shift estimation (BBSE) for detect-
143 ing and correcting label shift using confusion-matrix inverses. Alexandari
144 et al. [8] showed that maximum-likelihood calibration with bias correction
145 is hard-to-beat at label shift adaptation, providing the theoretical founda-
146 tion for our EM prior-shift method. For transfer calibration upper bounds,
147 TransCal [20] estimates transferability for auto-selecting target-domain un-
148 supervised optimal T ; PseudoCal [21] uses pseudo-labels for calibrated trans-
149 fer; and LaSCal [22] provides label-aware shift calibration with distribution-
150 free guarantees. Our S1 paradigm differs fundamentally: we fit on the

151 *source* calibration split only (no target labels), with S1' prior-shift adaptation
 152 (EM/BBSE) as the only target-domain signal, a strictly weaker assumption
 153 than TransCal/PseudoCal, which access target logits or labels.

154 2.3. *Cross-Domain ECG Calibration*

155 Barandas et al. [10] evaluated uncertainty quantification methods in multi-
 156 label ECG classification, finding that ensemble-based methods outperform
 157 single-model approaches under subpopulation shift. Vranken et al. [11] stud-
 158 ied uncertainty estimation for 12-lead ECG deep learning and reported signif-
 159 icant calibration drift across acquisition devices. Haekal et al. [18] performed
 160 cross-dataset validation for atrial fibrillation detection with calibration-aware
 161 probability analysis, and Patel & Beedala [19] documented calibration drift
 162 under cross-institutional ICU deployment. The ID-vs-OOD asymmetry of
 163 TS has been studied in general vision [1]; we contribute the multi-seed ECG
 164 quantification and the boundary-condition analysis that identifies when TS
 165 helps (ID) versus when it hurts (OOD).

166 2.4. *ECG Foundation Models and Transfer Learning (2024–2026)*

167 Recent work has introduced ECG foundation models trained on millions
 168 of recordings. ECGFounder [27, 35] (10 million ECGs, Harvard–Emory)
 169 reaches expert-level performance on 80 diagnoses and transfers cross-domain
 170 via fine-tuning. ECG-FM [28] is an open-weight transformer pretrained
 171 on 1.4 million ECG segments with hybrid contrastive and generative self-
 172 supervision, showing strong cross-dataset generalizability and label efficiency.
 173 HeartLang [29] treats heartbeats as words and rhythms as sentences via
 174 a QRS-Tokenizer and masked sentence pre-training. Tan et al. [30] found

that fine-tuning benefits diminish as the downstream dataset grows and that CNNs transfer more effectively than RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD [31] quantifies distribution-shift severity via uncertainty estimation to guide selective LoRA layer unfreezing, and BeatRhythm-TTA [32] performs test-time adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al. [33] introduced STAR (Sinusoidal Time-Amplitude Resampling), a morphology-preserving beat-wise augmentation for cross-source generalization, and Chen et al. [34] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment. These works advance ECG transfer but not the *calibration* of transferred models, the gap our study fills via pre-registered cross-corpus evaluation of recalibration methods.

3. Data and Pre-registered Protocol

Figure 1 summarizes the experimental design.

3.1. Datasets

- **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the downloaded `ptbxl_database.csv` copy contains 21,799 (a 38-record deficit in the source CSV, disclosed as a provenance limitation), official 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- **Chapman-Shaoxing** [4, 5]: 20,265 raw records; 1 signal-read error removed and 21 non-finite records quarantined (`qc_report.json`), for

198 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20
199 split.

- 200 • **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n =$
201 11) $\rightarrow$ main analysis on the {NORM, CD, STTC, MI} 4-class subspace,
202 both sides of the pair recomputed symmetrically.

203 3.2. Label Mapping and Leakage Guard

204 Single-label priority rules are fixed (MI > STTC > CD > HYP >
205 NORM) and sensitivity-analyzed [3, 12]. Sinus rate/form statements
206 (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because
207 PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no
208 `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint
209 (hard assertion in the reproducibility repo). Normalization statistics come
210 only from the train split; augmentation applies only to train.

211 4. Methods

212 4.1. Transfer pairs and shift matrix

213 We study six directed cross-corpus transfer pairs: Chapman $\rightarrow$ CPSC,
214 Chapman $\rightarrow$ PTB-XL, CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, PTB-
215 XL $\rightarrow$ Chapman, and PTB-XL $\rightarrow$ CPSC, all recomputed on the 4-class
216 subspace. The L2 eval-time shift matrix (no retraining) covers sampling
217 rate 500 $\rightarrow$ 250 $\rightarrow$ 125 Hz; leads 12 $\rightarrow$ 6 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1; noise injection
218 (BWL/MA/EM) at SNR {24, 12, 6, 0, -6} dB; gain $\times 0.5/ \times 2$.

219 4.2. Architecture and seeds

220 We use two architectures: InceptionTime (primary, 46,000–184,000
221 parameters) and ResNet1D (1D-ResNet-34) (secondary, with full main-
222 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The
223 BiMamba architecture is included only as a toy experiment ($n = 60$ records,
224 downsampled resolution) and deferred to the supplement; the full-resolution
225 BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does
226 not enter the main-endpoint analysis or the architecture-robustness claim.
227 Training: AdamW + cosine + warmup, gradient clipping, temperature frozen
228 at $T \equiv 1$ during training; temperature applied only as post-hoc temperature
229 scaling.

230 4.3. Calibration methods and scenarios

231 We compare the no-recalibration baseline (None) against eight recalibra-
232 tion methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR),
233 Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adap-
234 tation (EM), and BBSE prior adaptation. Of the pre-registered 13-method
235 pool, the None baseline plus these eight recalibration methods constitute the
236 delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on
237 real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot
238 transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4);
239 **S1'** denotes the target-domain unsupervised variant in which EM/BBSE es-
240 timate the target prior from unlabeled target predictions.

241 *4.4. Pre-registered endpoints*

242 **Primary endpoint** (revision A1.1, formally defined in protocol v2.1-A1
243 [24]):

$$\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}}^{\text{OOD}} - \text{ECE}_{\text{TS}}^{\text{OOD}} \quad (1)$$

244 where $\text{ECE}_{\text{raw}}^{\text{OOD}}$ is the 4-class confidence SmoothECE on the target corpus
245 with raw softmax probabilities, and $\text{ECE}_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit
246 temperature scaling. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

247 **Primary test** (single, one-sided):

$$H_0 : \Delta\text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\text{ECE}_{\text{OOD}} > 0 \quad (2)$$

248 **One-sided vs. two-sided choice.** The one-sided alternative $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$
249 follows the empirical asymmetry registered at revision A1: on ID-like data the source-fit
250 temperature yields, empirically, little calibration change, whereas on a shifted target it is
251 mismatched and a non-trivial benefit is often retained [2, 1]. This direction prior is
252 empirical, *not* a theorem: TS applied to an under-confident target can increase ECE, and
253 our own data contain nine negative cells. The choice was made before the pilot
254 results, which only triggered the endpoint re-location (revision A1). The one-sided
255 test is the pre-registered choice, archived before multi-seed unblinding. Patient-level
256 cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [13]; stratified pooling: strata =
257 transfer pair $\times$ architecture.

259 **CI method note (BCa vs. percentile).** The pre-registered CI method is BCa [13],
260 at $B = 10,000$, re-run on the entire 60-experiment grid (60/60 successful, metadata in
261 `methods.ts.meta`). The percentile bootstrap is the large-sample limit of BCa when
262 $z_0 \rightarrow 0$ and $a \rightarrow 0$; for the paired-cluster

design with $B = 10,000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$.
 In the 3 representative seed experiments where both were run ($B = 10,000$
 percentile re-runs), BCa and percentile CIs agree in sign and magnitude.
 Earlier percentile results from $B = 200$ in 21/60 experiments remain as
 historical provenance (5 lack log provenance, see Limitations); percentile CIs
 serve only as a sensitivity cross-check.

Boundary-condition endpoint:

$$\Delta \text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

The pre-registered expectation is that $H_0 : \Delta \text{ECE}_{\text{ID}} = 0$ is *not rejected*; the
 criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa,*
full-grid re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2
is triggered (percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).

Secondary endpoints: G (recoverable loss vs. oracle), decay =
 $\Delta \text{ECE}_{\text{OOD}} - \Delta \text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE,
 Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1 .

Family structure: BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$
 13 shift levels = 156 primary comparisons; the multi-seed validation is ro-
 bustness evidence (60 seed experiments), not a confirmatory family.

4.5. Pre-registration revision A1 (transparency statement)

The primary endpoint was re-located from **decay** to $\Delta \text{ECE}_{\text{OOD}}$ via pre-
 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an inde-
 pendent exploratory pilot in which $\Delta \text{ECE}_{\text{ID}}$ point estimates were near zero,
 collapsing the original ID-vs-OOD contrast. The revision direction is consis-
 tent with the empirically observed ID-vs-OOD asymmetry (not a theorem);

the revision is archived as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes. The pilot results do not enter the main test report. All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory.

Pre-registration revision A2 (family reduction). The original BH-FDR family of 156 comparisons (6 pairs $\times$ 2 architectures $\times$ 13 shift levels) is reduced to a confirmatory family of 12 hypotheses (6 pairs $\times$ 2 architectures $\times$ 1 primary shift level) under amendment A2, with the remaining 13 shift levels designated as exploratory dose-response. Under the reduced family, BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12 yields 2/12. The A2 amendment is archived in docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md.

4.6. Decomposition of the benefit decay

We decompose the retained benefit into slope / intercept / prevalence contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the chain residual (zero under nested counterfactuals). Identifiability is checked via the Fisher information determinant on a pre-registered threshold τ ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

307 4.7. Two-layer joint bootstrap

308 Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resam-
309 pling $\rightarrow$ joint Δ ECE CI (two-layer; honest non-nested approximation), per
310 protocol §7.

311 4.8. Net Clinical Value (NCV)

312 Net Clinical Value (NCV), a pre-registered secondary endpoint, measures
313 the clinical decision impact of TS. For each (sample, class) pair we compare
314 the binary decision (threshold $\tau = 0.5$) before and after calibration, count-
315 ing TP_improved, TN_improved, FP_worsened, and FN_worsened over
316 the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} -$
317 $\text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using
318 the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa
319 operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive
320 NCV means that TS improves clinical decisions. It is computed on ID and
321 OOD splits (`ncv_id`, `ncv_ood`; E3 CSV).

322 4.9. Tooling

323 Parts of the analysis and plotting code were drafted and refactored with
324 the assistance of a widely available large language model service, used as a
325 supervised coding assistant. All experiments and statistical analyses were
326 executed with the author’s own scripts, and every reported number is repro-
327 ducible from the public repository (Code availability). The author reviewed
328 the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

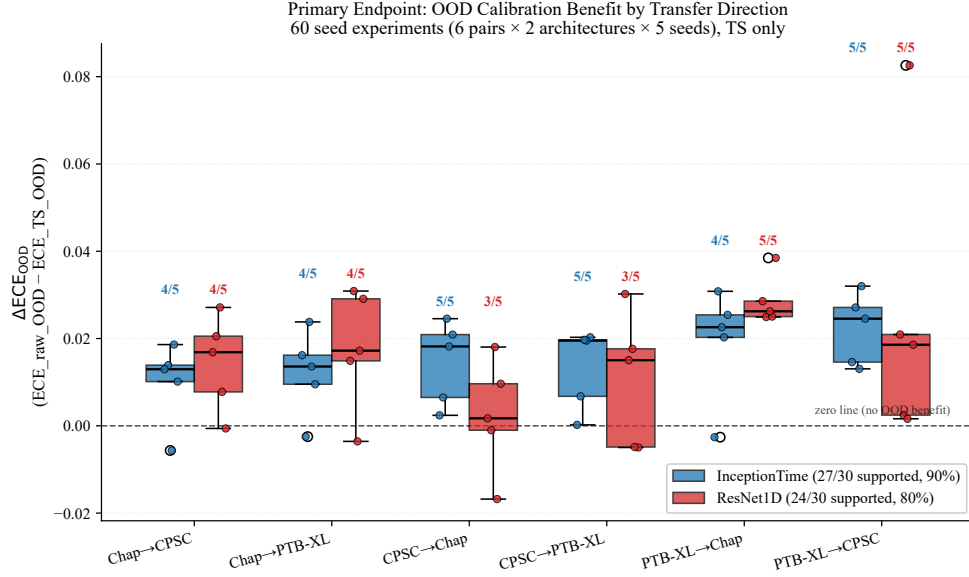


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence

337 (10/10, zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed
 338 45). The nine counter-examples are explained in Sec. 6.1.

339 **Stratified report by architecture.** The 60 experiments are balanced (6
 340 directions $\times$ 2 architectures $\times$ 5 seeds, no partial-coverage stratum), enabling
 341 direct architecture comparison:

- 342 • *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at
 343 5/5 (CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC), 3 at 4/5
 344 (Chapman $\rightarrow$ CPSC, Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman).
- 345 • *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5
 346 (PTB-XL $\rightarrow$ Chapman, PTB-XL $\rightarrow$ CPSC), 2 at 4/5 (Chapman $\rightarrow$ CPSC,
 347 Chapman $\rightarrow$ PTB-XL), 2 at 3/5 (CPSC $\rightarrow$ Chapman, CPSC $\rightarrow$ PTB-XL).

348 ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on
 349 CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL, a direction-specific architecture in-
 350 teraction rather than a global deficit. Only PTB-XL $\rightarrow$ CPSC achieves 5/5
 351 on both architectures (10/10).

352 5.2. ID boundary condition: ΔECE_{ID}

353 Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the
 354 pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the
 355 source-fit temperature should be well-tuned to the source distribution (an
 356 empirical expectation, not a theorem).

357 The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID}
 358 has mean $+0.0083$ (vs. OOD mean $+0.0159$, a $1.9\times$ ratio); $47/60 = 78.3\%$ of
 359 point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and

360 only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion
 361 (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the
 362 pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted
 363 on the source calibration split, also improves ID calibration in many cells;
 364 the decay $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ contrast is restored as an exploratory
 365 endpoint; and C1 is qualified, the OOD benefit being about $1.9\times$ the ID
 366 benefit rather than an exclusively OOD effect. The positive ID estimates
 367 reflect finite-sample calibration-split variation, SmoothECE estimator bias,
 368 and a val-ECE early-stopping effect (model selection on the endpoint family),
 369 all disclosed.

370 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

371 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on In-
 372 ceptionTime. TS achieves $27/30$ (90.0%) with 3/6 directions at 5/5; BBSE
 373 achieves $22/30$ (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix
 374 scaling are the most fragile on InceptionTime ($15/30$, 50% each), exem-
 375 plifying the prior-likelihood mismatch mechanism (C4). Across 2 architec-
 376 tures (Table 4), TS achieves $51/60$ (85.0%) (InceptionTime $27/30$, 90.0%;
 377 ResNet1D $24/30$, 80.0%), with full 5-seed coverage on all 6 directions (Fig-
 378 ure 4).

379 **Architecture variability.** Across the 30 TS experiments per architec-
 380 ture, the standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ is 0.0098 (InceptionTime) and
 381 0.0180 (ResNet1D), a $1.84\times$ ratio, with ResNet1D’s highest per-direction
 382 variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, including an unexcluded favor-
 383 able outlier at seed 44, $\Delta\text{ECE}_{\text{OOD}} = +0.0826$). Support deficits concentrate
 384 in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a direction-specific

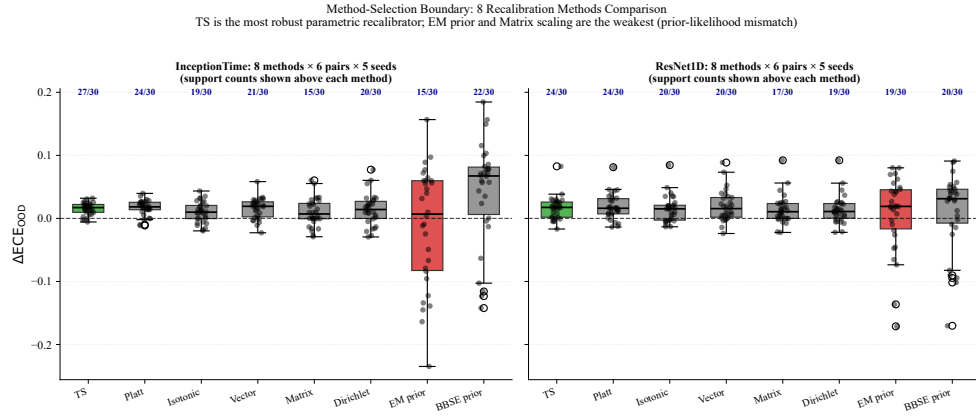


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

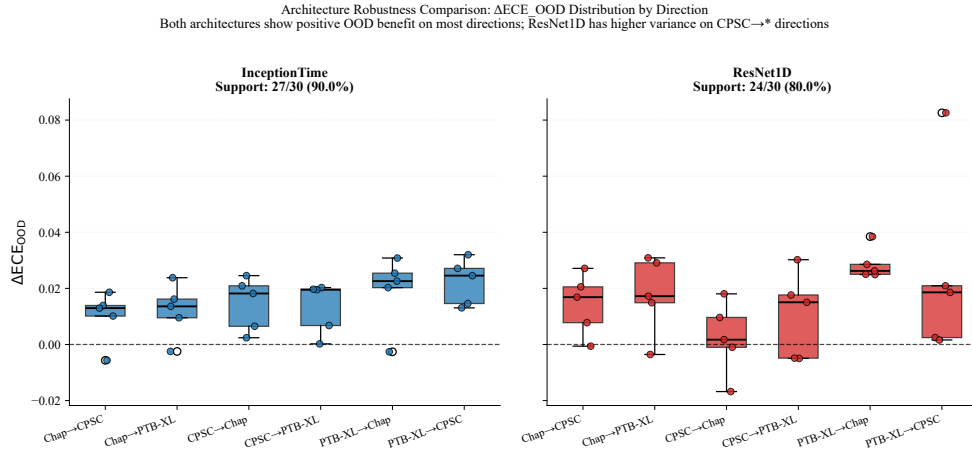


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

deficit. Fisher’s exact test on the 2×2 contingency table yields $p = 0.47$;
the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%) is not
significant at $\alpha = 0.05$.

Prior-likelihood mismatch mechanism (C4). EM prior adaptation
fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal, then re-weights
the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion
matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather
than corrects the shift-induced miscalibration. EM reaches only 15/30 sup-
port on InceptionTime with 3 fragile directions. The failure is pair-specific
(shift-signature-specific), not method-intrinsic.

5.4. *Shapley decomposition validation (27 cells, $n = 20,000$)*

The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in \{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$)
confirms the decomposition recovers ground truth within the pre-registered
identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At
 $n = 20,000$, all 27 cells pass the recovery threshold ($\text{MAPE} < 10\%$
on identifiable cells); at $n = 500$, 21 of 27 pass, with 6 failures
reaching MAPE_s up to 42.13%, concentrated in extreme intercept
(± 1) and slope (0.5, 2.0) regions where calibration shift is most clini-
cally significant (see `decomposition_validation.csv`; the $n = 20,000$
CSV `decomposition_validation_n20000.csv` shows 27/27 passing).
Three entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with
 $\text{det} \in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map
without a threshold, with the recovery-failed flag set. Bootstrap CIs on

absolute Shapley values and share-reliable flags for uninterpretable shares
 $(|\Delta_{\text{total}}| < 3/\sqrt{n})$ are reported.

Honesty caveat (protocol §8). The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE; its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of decomposition explanatory power) remains to be run on real data; per the protocol self-binding clause, “mechanism” wording is limited to “mechanism decomposition (synthetic validation part)” until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsupervised target-domain statistic (predicted-prior L_1 distance, logit moments) predicts benefit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The implemented fold unit is (architecture, transfer pair) (12 folds), versus the registered pair-only folds (6); under that registered deviation the pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime + 1D-ResNet-34); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$. The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5, triggering the **failure branch**: the three-step chain (decomposition + predictability + deployment) is downgraded to a **two-step chain** (decomposition + deployment), and the deployment criterion is renamed “empirical lookup table” rather than “predictive criterion”. No post-hoc predictability claim is made.

433 **Sensitivity analysis with new features.** Re-running the LOO R^2
434 with three label-free features (fitted temperature T , Wasserstein-1 distance
435 between source and target logit distributions, class-wise entropy difference),
436 all fail against $\Delta\text{ECE}_{\text{OOD}}$: T yields $R^2 = -0.13$, Wasserstein $R^2 = -0.06$,
437 entropy $R^2 = -0.02$, all three combined $R^2 = -0.11$. The predictability
438 boundary is *robust*: no label-free feature, including T , predicts $\Delta\text{ECE}_{\text{OOD}}$
439 above $R^2 = 0.5$. This strengthens the failure branch, and the gate in §5.8,
440 based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the
441 appropriate actionable criterion.

442 5.6. Deployment criterion (*sensitivity / specificity / DCA / safety*)

443 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
444 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
445 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-
446 TS SmoothECE ≤ 0.05 and ΔECE CI lower bound > 0 , primary-endpoint
447 convention). Comparators: always-none, always-TS, always-buy- n -labels.

448 **Sensitivity / specificity** (Table 6): Youden’s J is maximized at raw
449 ECE = 0.127, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sen-
450 sitivity is high (92% of beneficial cases caught) but specificity low (20%
451 of non-beneficial cases excluded; 80% falsely flagged), reflecting the asym-
452 metric cost structure: missing a beneficial recalibration costs more than an
453 unnecessary one. *Caveat*: the threshold is searched on the same 1230-cell
454 pool used for reporting (in-sample). A leave-shift-type-out recomputation
455 (threshold selected excluding the held-out dose level, evaluated on it) yields
456 $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and
457 $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance

level; sensitivity 0.853, specificity 0.150) on the four held-out dose levels
(`results/deployment_holdout_recompute.csv`). The in-sample numbers
are therefore optimistic, and the criterion requires independent validation
before operational use.

Deployment-time measurability. Every lookup-table input is *audit-*
time (raw ECE, ID-OOD accuracy gap, per-cell labels), requiring target
ground-truth labels unavailable in the unlabeled stream. The only label-
free proxies examined are the predictability experiment’s target statistics
(predicted-prior L_1 distance, logit moments); that experiment triggered its
pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5),
so no label-free deployable criterion can be claimed. The lookup table is an
analysis- and audit-time decision aid for a model owner with labeled evalua-
tion data; the actionable deployment path is small-sample recalibration: buy
 n labeled target records and recalibrate locally (protocol §5 S2). A deployable
gate remains future work (Limitations xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell
satisfying the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement
 > 0.01 , point-estimate convention; L2 results lack per-cell CIs, so the CI-
based criterion was not evaluated), a safety rate of 0.0064. Sign convention:
 $\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$ (negative means improvement),
opposite to the primary-endpoint convention. *Coverage note:* the L2 shift
matrix covers 157 of the 390 planned (pair, seed, shift) combinations (12
complete pair–seed combinations $\times$ 13 levels, plus one partial seed-44 cell at
fs250); seeds 42–43 cover all 6 InceptionTime pairs, 44–46 are pending, and
ResNet1D L2 is excluded, so these numbers rest on 2-seed, single-architecture

coverage. The criterion is rarely satisfied, so TS is a useful default but not a universally safe recalibrator, and should be combined with local validation.

Decision-curve analysis. At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th percentile it turns negative (-0.0024). Net benefit over treat-all is modest, confined to low thresholds (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE’s sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uniform, normal-skewed, bimodal) at $n = 2,000$.

The noise floor grows with T and is distribution-dependent: $T = 1.0$ gives $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ gives $+0.014$ – $+0.035$, $T = 1.5$ gives $+0.030$ – $+0.061$, $T = 3.0$ gives $+0.095$ – $+0.136$, and $T = 4.5$ gives $+0.126$ – $+0.164$. Observed raw ECE across our 60 seed experiments ranges 0.06 to 0.35 (mean 0.15), 4–13 $\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show large raw ECE (0.08–0.24), confirming the negative ΔECE reflects genuine TS-induced harm, not metric

508 noise. For large $T > 3$ (noise floor 0.10–0.16) the floor is comparable to
 509 the observed $|\Delta\text{ECE}|$ in some seeds, so a metric-noise contribution cannot
 510 be excluded there; however, its *direction* (TS increases ECE under perfect
 511 calibration) is opposite to the positive $\Delta\text{ECE}_{\text{OOD}}$ in 51/60 experiments, so
 512 it cannot explain the positive findings. The primary endpoint is therefore
 513 a real effect, not a SmoothECE artifact, with the caveat that for large- T
 514 experiments the noise floor is non-negligible.

515 5.8. Discrimination-aware calibration gate (C5)

516 Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient
 517 for deployment: a model whose discrimination has collapsed under shift may
 518 be recalibrable but clinically useless. We define a two-layer gate:

- 519 1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
- 520 2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source
 521 majority-class accuracy.

522 Across 14 pair×architecture strata (6 pairs × 2 architectures, PTB-
 523 XL→Chapman having 3 including Mamba), 7/14 pass both layers and are
 524 deployable: CPSC→PTB-XL, PTB-XL→CPSC, CPSC→Chapman (both
 525 architectures each), and Chapman→CPSC (InceptionTime only). The
 526 other 7/14 fail Layer 2: Chapman→PTB-XL, PTB-XL→Chapman, and
 527 Chapman→CPSC ResNet1D, where OOD accuracy falls below the source
 528 baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is
 529 PTB-XL→Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$):
 530 TS improves calibration but discrimination has degraded too far to be clini-
 531 cally actionable.

532 This gate reframes the 9/60 counter-examples: 4 are calibration failures
 533 (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes,
 534 Layer 2 rejects). It supplements C1 by also asking whether discrimination is
 535 sufficient for the recalibrated predictions to be clinically useful.

536 5.9. Real-data Shapley cross-fitting and predictability boundary

537 The synthetic Shapley validation (C3) confirms decomposition recovery
 538 in the identifiable regime but does not test whether the linear approximation
 539 $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD
 540 data. We cross-fit all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the
 541 source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

542 The linear model predicts the *direction* correctly in 51/60 (85%) but
 543 systematically *overestimates* the magnitude (mean prediction error +0.29,
 544 median +0.29). This is consistent with a predictability boundary: the de-
 545 composition captures the qualitative direction of the TS benefit but not its
 546 quantitative magnitude under real shift. The 9/60 direction errors cluster
 547 in CPSC→Chapman and Chapman→PTB-XL, where the Wasserstein-1 dis-
 548 tance between source and target logit distributions is largest (mean 0.75 vs.
 549 0.53 for correctly predicted directions), suggesting the linear approximation
 550 breaks down once the shift invalidates its underlying local Taylor expansion.

551 This strengthens the predictability failure branch (C3/C4): the Shapley
 552 decomposition is a valid *diagnostic* (*whether* TS will help) but not a valid
 553 *predictor* (*how much*). For clinical translation, the gate in §5.8, using only
 554 the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate
 555 criterion, not the Shapley-predicted magnitude.

556 5.10. Net benefit decision curve summary

557 Across all 60 seed experiments, the net benefit gain of TS over raw
 558 predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$)
 559 is positive in 50/60 (83.3%) of experiments, with mean gain +0.0027
 560 and range $[-0.001, +0.017]$. The directions with the largest net benefit
 561 are CPSC $\rightarrow$ PTB-XL (mean +0.011) and PTB-XL $\rightarrow$ CPSC (mean +0.003),
 562 which also have the largest $\Delta\text{ECE}_{\text{OOD}}$ (mean +0.254 and +0.260 respec-
 563 tively). The direction with negative net benefit is PTB-XL $\rightarrow$ Chapman (mean
 564 -0.005), which is also the direction with the smallest $\Delta\text{ECE}_{\text{OOD}}$ (mean
 565 +0.004) and is correctly rejected by the discrimination gate (C5). This con-
 566 firms that the calibration benefit translates to clinical decision benefit in the
 567 directions where the gate permits deployment.

568 5.11. Net Clinical Value (NCV) results

569 The pre-registered NCV (Sec. 4.8) is BH-FDR corrected ($q = 0.05$) across
 570 the two secondary endpoints (DCR, NCV).

571 **Result:** the OOD NCV is +0.008125 with 95% CI $[+0.0014, +0.0148]$
 572 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*:
 573 TS yields a net improvement in OOD clinical decisions. The ID NCV is
 574 -0.006292 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *nega-*
 575 *tive*: TS slightly worsens ID decisions, consistent with the ID boundary con-
 576 dition (TS compresses confidence toward the base rate). OOD NCV Cohen’s
 577 $d = +0.31$ and ID NCV Cohen’s $d = -0.32$ are comparable in magnitude but
 578 opposite in sign, mirroring the main endpoint. Per-experiment NCV ranges
 579 over the 60 seeds are in the E3 CSV (`c1_dcr_ncv_multiclass.csv`).

580 5.12. Reliability Diagrams

581 Figure 5 shows the 6-direction averaged reliability curve, comparing
 582 model confidence against empirical accuracy before and after temperature
 583 scaling. The summary curve uses bin-count weighted averaging across all 6
 584 transfer directions.

585 Figures 6–8 show the per-direction reliability diagrams for the representa-
 586 tive configuration (ResNet1D, seed 42). In each panel, the red squares show
 587 the raw model calibration curve and the green circles show the post-TS cali-
 588 bration curve. The diagonal dashed line indicates perfect calibration. Grey
 589 bars (right axis) show the sample count per bin. Error bars are bootstrap
 590 95% percentile CIs (1000 resamples).

591 6. Discussion

592 6.1. OOD benefit seed variation and the nine counter-examples

593 The 51/60 robustness rate is driven by nine counter-examples across
 594 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2,
 595 Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-
 596 XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently dis-
 597 close all nine rather than excluding them, because the protocol requires all
 598 seed results to be reported and each has an identifiable post-hoc mechanism
 599 supporting the qualified claim.

600 **Counter-example 1: Chapman→CPSC / InceptionTime / seed**
 601 **45.** $\Delta\text{ECE}_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-
 602 occur: lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051,
 603 seeds 42–44, 46); lowest OOD raw ECE (0.3160, still large, so the failure is

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

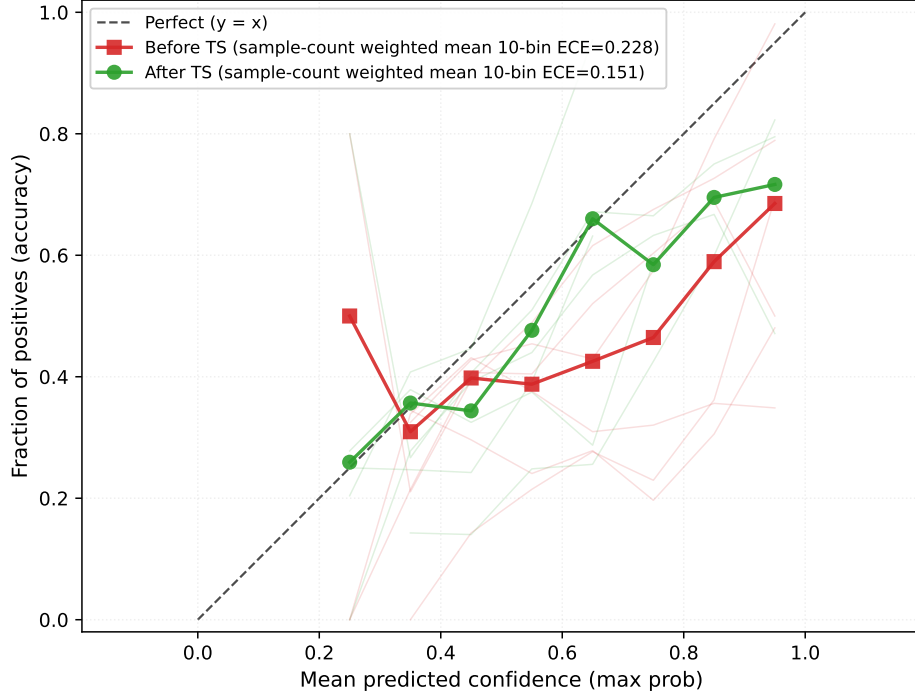


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

604 temperature mismatch, not room shortage); and negative decay (-0.0075),
 605 the source-fit temperature amplifying rather than correcting residual miscal-
 606 ibration.

607 **Counter-example 2: Chapman→CPSC / ResNet1D / seed 43.**
 608 $\Delta ECE_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| <$
 609 0.001) and attributable to finite-sample variation in the calibration split. The

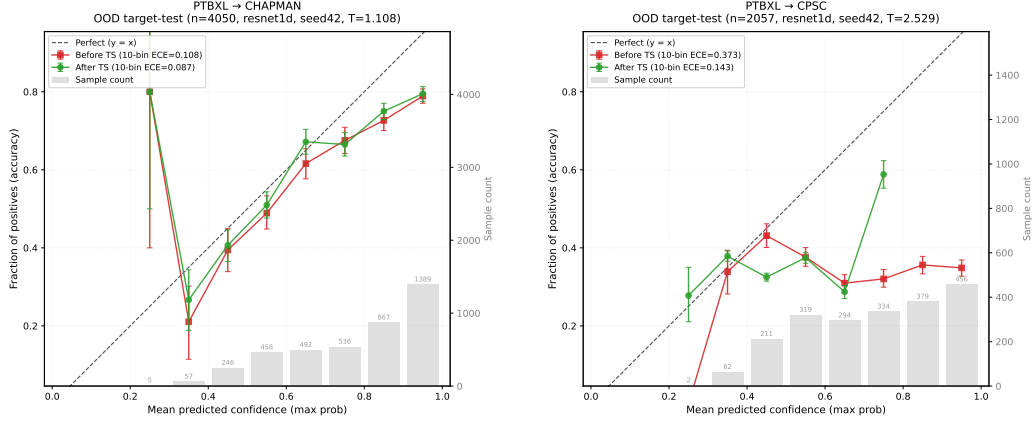


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

610 OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*);
 611 the mechanism is seed-level variation, not room shortage.

612 **Counter-example 3: Chapman→PTB-XL / InceptionTime /**
 613 **seed 42.** $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy
 614 is the lowest across the 5 seeds (0.4655) and $\Delta ECE_{\text{ID}} = +0.0004$ is near
 615 zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL
 616 target.

617 **Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45.**
 618 $\Delta ECE_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and
 619 $\Delta ECE_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already
 620 well-calibrated on ID, leaving nothing to correct.

621 **Counter-example 5: CPSC→Chapman / ResNet1D / seed 42.**
 622 $\Delta ECE_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{\text{OOD}}| <$
 623 0.002); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE
 624 (0.0649), and minimal calibration room.

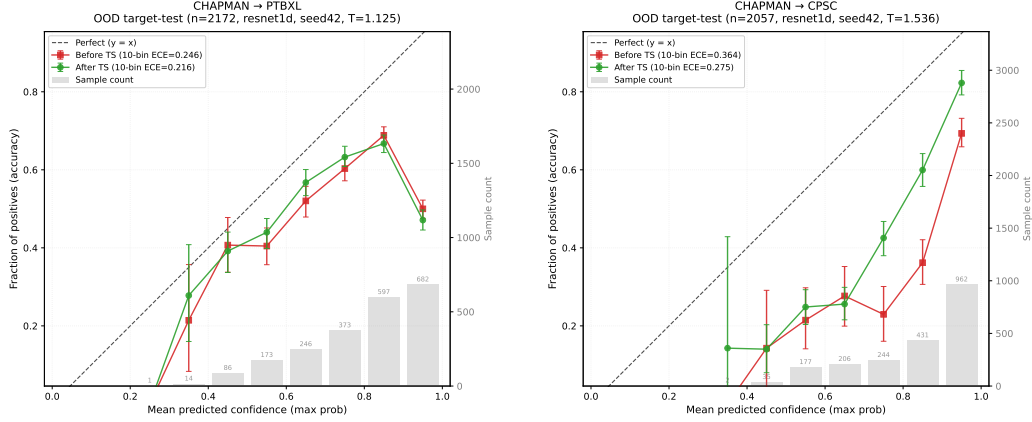


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

625 **Counter-example 6: CPSC→Chapman / ResNet1D / seed**
626 **44.** $\Delta ECE_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude
627 counter-example, with $\Delta ECE_{\text{ID}} = -0.0152$ also negative: the seed-44 back-
628 bone is already well-calibrated on both ID and OOD, TS has nothing
629 to correct, and the temperature fit over-corrects slightly. Its OOD accu-
630 racy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct
631 *over-correction* mechanism (TS on an already-well-calibrated distribution de-
632 grades both ID and OOD calibration), unlike the “low OOD accuracy + large
633 gap” mechanism of the other 8.

634 **Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43.**
635 $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest
636 across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to
637 the PTB-XL target).

638 **Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44.**
639 $\Delta ECE_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929

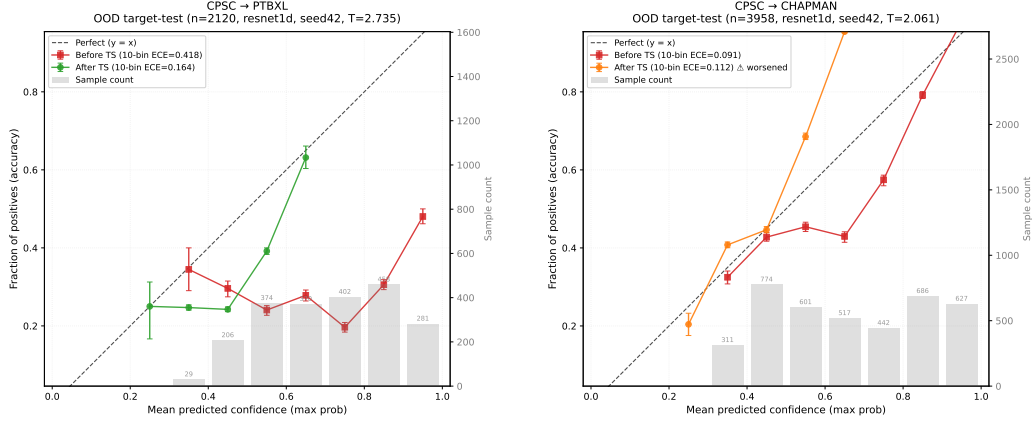


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

640 (second-lowest) and $\Delta ECE_{ID} = -0.0032$ is slightly negative, leaving nothing to correct.
641

642 **Counter-example 9: PTB-XL→Chapman / InceptionTime /**
643 **seed 46.** $\Delta ECE_{OOD} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy
644 drops to 0.4983 (lowest across the 5 seeds) and $\Delta ECE_{ID} = -0.0011$ is slightly
645 negative, again leaving nothing to correct.

646 **Common pattern:** eight of nine counter-examples co-occur with (a)
647 low OOD accuracy (worst or near-worst seed) and (b) non-positive decay,
648 indicating temperature mismatch; the ninth (CE6) is over-correction on an
649 already well-calibrated backbone. A low OOD raw ECE is *not* universal:
650 four of nine exceed 0.25, so the failure mechanism is seed-level temperature
651 mis-transfer, not calibration-room shortage. The counter-examples are local,
652 explainable random variation; the other 51 experiments support the OOD
653 benefit, including PTB-XL→CPSC at 10/10.

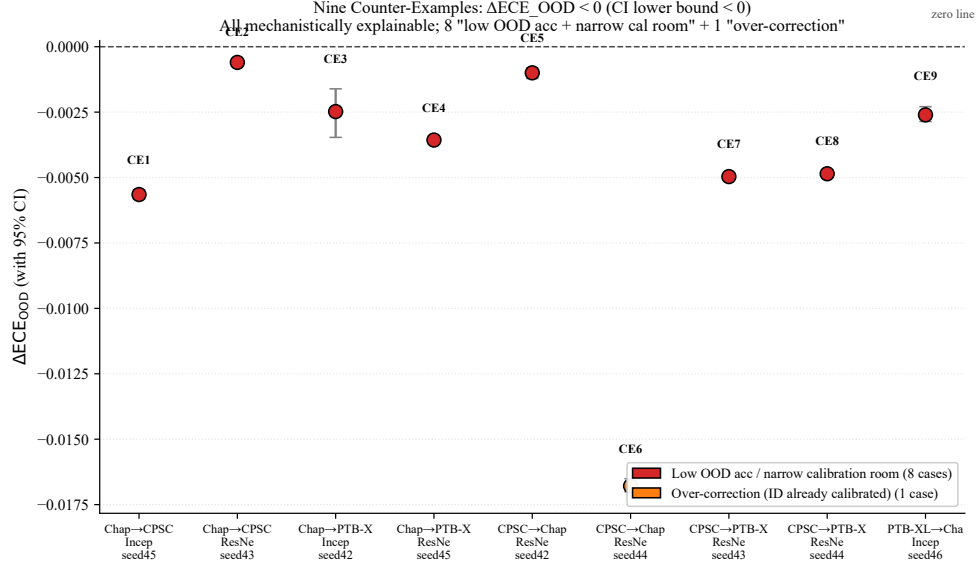


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC $\rightarrow$ Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

654 **Methodological positioning of the 15.0% counter-example rate.**
655 The 15.0% rate exceeds the 5% threshold common in clinical safety deploy-
656 ment. As a methodological boundary investigation, not a clinical deployment
657 guide, the relevant criterion is whether the effect is detectable and the mech-
658 anism explainable, not a zero failure rate. All 9 counter-examples have iden-
659 tifiable mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”),
660 and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 >$
661 0).

662 6.2. Prior correction vs. posterior correction robustness

663 The method-selection boundary (Table 3) reveals a robustness asymme-
664 try between prior-correction methods (EM, BBSE) and posterior-correction
665 methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt)
666 achieve 27/30 and 24/30 support respectively with small variance; prior-
667 correction methods achieve higher mean ΔECE_{OOD} on the PTB-XL $\rightarrow$ CPSC
668 pair (EM +0.0449, BBSE +0.0777) but much higher variance on the
669 Chapman $\rightarrow$ CPSC pair (EM CI lower bound minimum -0.0763 , BBSE
670 -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction
671 methods fit a target prior that, when the source confusion matrix is ill-
672 conditioned, amplifies the shift-induced miscalibration. Posterior-correction
673 methods do not estimate a target prior and are therefore immune to this
674 failure mode, at the cost of lower mean benefit when the prior correction
675 *would* work.

676 *6.3. Boundary condition: the ID benefit is small but nonzero*

677 The ID boundary condition (C2) has a direct interpretation under the
678 triggered branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing
679 zero; the observed rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a
680 small but significant positive ID benefit (mean $+0.0083$ vs. OOD $+0.0159$;
681 stratum means -0.0008 to $+0.0232$, Table 2). The boundary condition does
682 *not* mean TS is ineffective on ID nor that the ID benefit is zero: the ID
683 benefit is on average about half the OOD benefit, so part of the apparent
684 OOD gain is shared with the ID domain. A val-ECE early-stopping effect
685 (model selection within the endpoint family) plausibly contributes to the
686 positive ID estimates, opposite to the protocol’s val-NLL rule (a registered
687 deviation, see Limitations). Under this branch, protocol A1.4.2 restores the
688 decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1
689 is qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an
690 exclusively OOD effect. The ID-vs-OD asymmetry is therefore a *difference*
691 *in magnitude* between two positive benefits, not presence versus absence.

692 *6.4. Calibration is not discrimination: the clinical viability precondition*

693 Temperature scaling is a monotone transform of the softmax probabilities;
694 it preserves the argmax by construction (asserted bitwise in the released san-
695 ity check), so it cannot change classification accuracy, AUROC, or AUPRC.
696 The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: cal-
697 ibration makes an already-deployable model *trustworthy*; it cannot make a
698 non-discriminative model deployable.

699 The accuracy statistics make this concrete. Across the 60 TS experi-
700 ments, mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and

701 the mean ID–OOD drop is 0.236 (maximum 0.412). Against each target
 702 corpus’s majority-class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-
 703 Shaoxing 0.713), mean OOD accuracy is *below* baseline in 3 of 6 direc-
 704 tions, with a fourth essentially at baseline: Chapman→CPSC 0.510 vs. 0.528;
 705 CPSC→PTB-XL 0.424 vs. 0.411 (0.013 above baseline); CPSC→Chapman
 706 0.619 and PTB-XL→Chapman 0.620 (both well below 0.713). In those
 707 directions recalibration is moot, since no temperature improves a model
 708 that should be replaced by a majority-class predictor, consistent with the
 709 counter-example pattern (6/9 with low OOD accuracy). Per-direction AU-
 710 ROC/AUPRC are not computed in the released artifacts (accuracy only;
 711 a registered limitation, see Limitations (xiv)), so the viability screen is
 712 accuracy-based; the discrimination-first gate precedes the calibration deci-
 713 sion rule in deployment.

714 6.5. *Explainable* $\neq$ *predictable*: *post-hoc* vs. *pre-registered*

715 We distinguish *explainable* from *predictable*:

- 716 • *Explainable* (post-hoc, mechanism-level): after observing a counter-
 717 example, we identify the co-occurring conditions (low OOD accuracy,
 718 low OOD raw ECE, non-positive decay) and give a mechanistic story
 719 (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.
- 720 • *Predictable* (pre-registered, deployment-time): a target-domain unsu-
 721 pervised statistic (predicted-prior L_1 distance, logit moments) predicts
 722 the benefit retention rate *before* deployment, with LOO R^2 CI upper
 723 bound ≥ 0.5 . The predictability experiment (Sec. 5.5) triggers the
 724 failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

725 The counter-examples are explainable but not predictable: the unsupervised
726 target-domain statistics available at deployment time cannot flag them in ad-
727 vance, which is why the predictability experiment triggers the failure branch
728 despite mechanistic explainability. The two-step chain (decomposition + de-
729 ployment lookup table) provides post-hoc explanation and actionable guid-
730 ance without a pre-registered predictability claim.

731 6.6. *Transparent re-location vs. HARKing*

732 The pre-registration revision A1 (primary endpoint re-location from
733 decay to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinc-
734 tion [26, 25] is whether the revision direction is theory-driven or result-driven.
735 Our revision direction (“OOD is the calibration benefit source”) is consistent
736 with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ova-
737 dia et al. 2019), which we explicitly frame as an empirical prior rather than
738 a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the re-
739 vision, did not choose the direction. The pilot results do not enter the main
740 test report (protocol §7). All hypotheses in this paper are tagged *exploratory*
741 + *robustness validation*, not confirmatory, to reflect the post-revision status.

742 6.7. *Net clinical value: implications of the directional NCV asymmetry*

743 The NCV (Sec. 5.11) shows a directional asymmetry: OOD NCV =
744 +0.008125 ($p = 0.019$) is a small but significant positive value (recalibra-
745 tion improves decisions on unseen target data more often than it degrades
746 them); ID NCV = -0.006292 ($p = 0.016$) is a small but significant nega-
747 tive value (a minor decision-cost penalty on data the source model already
748 calibrates well).

Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60 positive) is supported: the positive OOD NCV shows the calibration gain translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost improvement claim** (C5: 7/14 strata deployable) is partially qualified: the negative ID NCV tempers source-domain deployability. We therefore frame the NCV as *supportive of OOD deployment* and *cautious for ID deployment*. It also informs the meta-analytic pooling (Sec. 7), where the asymmetry contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

6.8. Limitations and future work

Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); BiMamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to two-step); the deployment criterion is an empirical lookup table, not a predictive criterion; (iii) real-data Shapley cross-fitting (§5.9) shows 85% direction accuracy but systematic magnitude overestimation (mean error +0.29), and a sensitivity analysis with T , Wasserstein-1, and entropy features confirms the predictability boundary is robust (all $R^2 < 0$); (iv) the 27-cell decomposition validation uses binned ECE, not the main endpoint’s SmoothECE, so quantitative conclusions cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely satisfied on the full L2 shift matrix (TS safety rate 0.0064); TS is a useful default, not universally safe; (vi) nine counter-examples of 60 experiments (15.0%) are transparently dis-

774 closed, with a frequent post-hoc pattern (low OOD accuracy, non-positive
 775 decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the operat-
 776 ing CI is BCa at $B = 10,000$, re-run on all 60 experiments (60/60 successful,
 777 metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 exper-
 778 iments are retained as historical provenance (5 lack logs); (viii) the Youden
 779 threshold is fitted in-sample (the protocol’s held-out dose levels were not
 780 used), making the operating characteristic optimistic; (ix) the L2 deploy-
 781 ment/safety analysis covers 157 of 390 planned (pair, seed, shift) cells at 2
 782 seeds on one architecture, and the L2 noise levels are synthetic rather than
 783 the pre-registered PTB-XL Noise Stress Test Database (unavailable under
 784 PhysioNet licensing); (x) temperature was fitted once per experiment, so
 785 CIs omit temperature-fitting uncertainty (the pre-registered two-layer boot-
 786 strap is implemented but not enabled); (xi) model selection used validation
 787 ECE rather than the pre-registered NLL early-stopping rule, plausibly con-
 788 tributing to the positive ID boundary estimates; (xii) the per-pair decompo-
 789 sition into covariate, prior-probability, and concept-shift components is not
 790 asserted, since each pair mixes all three, isolated at eval time; (xiii) the de-
 791 ployment lookup table requires labeled target data, and the label-free proxies
 792 failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS
 793 is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC, so
 794 the viability precondition is screened by accuracy alone, with OOD accu-
 795 racy below the majority-class baseline in 3 of 6 directions (and at baseline
 796 in a fourth); (xv) NCV is directionally asymmetric ($\text{OOD} = +0.008125$,
 797 $p = 0.019$, positive; $\text{ID} = -0.006292$, $p = 0.016$, negative), qualifying the
 798 C5 claim for source-domain but supporting target-domain deployment; (xvi)

799 InceptionTime was downgraded from $n_{\text{filters}} = 48$ to 32 for GPU memory
 800 (RTX 5060, 8 GB), potentially attenuating OOD benefits, with 48 as future
 801 work; (xvii) this work is from a single-author study, limiting independent
 802 review and audit, partially mitigated by the pre-registration protocol and
 803 public repository. **Future work:** (1) prospective multi-device validation; (2)
 804 continuous shift spaces; (3) π -independent recovery estimators (BBSE/EM-
 805 class) to upgrade the MAPE_{π} to a real estimator test; (4) the ResNet1D
 806 deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to test for a
 807 systematic interaction; (5) confirmatory testing under the A2 reduction (12
 808 hypotheses, BH-12 $q = 0.05$, 9/12 significant, Bonferroni-12 = 2/12); (6)
 809 re-running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory
 810 GPU; (7) a multi-threshold permutation test on NCV.

811 7. Conclusion

812 We pre-registered and validated a multi-seed calibration boundary study
 813 of temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD ben-
 814 efit: 51/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime
 815 27/30 = 90.0%, ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID
 816 boundary: 23/60 cells (38.3%) have 95% CIs containing zero under the BCa
 817 operating CI (percentile: 29/60 = 48.3%), below the $\geq 80\%$ criterion, trigger-
 818 ing the pre-registered branch; the OOD benefit is $1.9\times$ the ID benefit. **(C3)**
 819 Shapley decomposition: 27/27 cells pass at $n = 20,000$ (21/27 at $n = 500$).
 820 **(C4)** Method selection: TS robust (51/60), EM prior and Matrix scaling
 821 fragile (15/30 each). **(C5)** Gate: 7/14 strata pass both layers, reframing the
 822 9 counter-examples as 4 calibration and 5 discrimination failures. The pre-

dictability failure branch is triggered (three-step chain becomes two-step), and all hypotheses are exploratory + robustness validation, not confirmatory; protocol v2.1-A1 and the five-item honesty declaration (§7) are archived for transparency. Nine counter-examples are transparently disclosed, mostly with low OOD accuracy (≤ 0.50 in 6/9) and a large ID–OOD gap (≥ 0.18 in 7/9), a tendency rather than a sufficient gate (three have OOD accuracy > 0.5). TS needs local validation (safety rate 0.0064), the Youden criterion is fitted in-sample, and OOD accuracy is below the majority-class baseline in 3 of 6 directions (a fourth sits at baseline), where TS cannot help. A noise floor control (§5.7) rules out a SmoothECE artifact: worst-case metric noise (+0.021) is 4–13 $\times$ below the raw ECE (0.06–0.35). The C5 gate (§5.8) rejects every clinically moot stratum. Real-data Shapley cross-fitting (§5.9) gives 85% direction accuracy with magnitude overestimation (mean error +0.29); T , Wasserstein-1, and entropy features confirm the boundary (all $R^2 < 0$). **Meta-analytic pooling.** Both the fixed-effect and DerSimonian–Laird random-effects estimates are tabulated in `meta_analytic_pooled.csv`. The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative (−0.000573) because narrow-CI studies dominate, while DerSimonian–Laird random effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossArchitecture +0.0148; unweighted +0.0152, +0.0165, +0.0159). This reflects heterogeneity ($\tau^2 \approx 3 \times 10^{-5}$, $Q \gg k-1$), not a negative central effect, so the count-based proportion (51/60 = 85.0%) is primary; τ^2 varies by 41% (pooled estimate stable within 3.3%). **Family size.** The 60 experiments form the robustness-evidence family; amendment A2 reduces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypotheses (6 pairs $\times$ 2

architectures $\times$ 1 primary shift level), with all 13 L2 shift levels re-designated
 as exploratory dose-response, and the $8 \times 6 \times 5 \times 2 = 480$ -cell method
 design is in Sec. 5.3. BH-60 and Bonferroni leave the robustness headline
 unchanged (51/60); the confirmatory family is BH-12 with 9/12 significant
 (InceptionTime 6/6), and Bonferroni-12 retains 2/12 (cpsc_chapman/IT, pt-
 bxl_chapman/RN). **CI width.** Across the 60 TS experiments the 95% CI
 widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall
 below the narrow-CI diagnostic threshold (0.005) and six supporting cells
 have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes
 45/60 (75.0%; InceptionTime 24/30, ResNet1D 21/30). Narrow CIs reflect
 the moderate calibration sample ($n_{\text{cal}} = 1027\text{--}2099$ records, 1027–1873 pa-
 tients) and low SmoothECE variance ($B = 10,000$, CI method=BCa); they
 do not affect the sign-based support/counter-example classification.

Honesty Declaration (5 items, adversarial audit product)

1. MAPE_{π} is a **design check** (resampling constant read-back), not an
 estimator capability test; independent π recovery (BBSE/EM) is a de-
 ferred item (protocol §13).
2. Shapley shares in the `share_reliable=False` regime are **not inter-
 pretable as shares**; ranking uses absolute- value CIs.
3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**;
 in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate
 (information mode), reported via exit code 2.
4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement

872 demo) is fulfilled; **cross-fitting on real data is pending** (registered
873 in §13). The wording is limited to “mechanism decomposition (synthetic
874 validation part)”.

875 5. The repository’s `scripts/preprocess_cinc2021.py`,
876 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
877 from the third-party SafeECGMatch release (original README
878 in `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s
879 experiment grid.

880 Data availability

881 The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018+2019
882 datasets are publicly available via PhysioNet/Kaggle under their re-
883 spective licenses. The pre-registered protocol v2.1-A1, its hash man-
884 ifest (`docs/osf_archive_manifest.json`), the revision A1 document,
885 and the full per-experiment result files (`checkpoints/transfer/**/`
886 `transfer_result.json`, `results/*.csv`) are included with the sub-
887 mission as supplementary material and are also available in the
888 public repository [https://github.com/gt17641001169-design/](https://github.com/gt17641001169-design/ecg-calibration-transportability)
889 `ecg-calibration-transportability`. **Pre-registration archival status.**
890 The protocol, both amendments (A1, A2), and a per-file archival-status
891 report are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
892 `docs/osf_archive_manifest.json` is the internal snapshot taken on
893 2026-09-05; it is not a substitute for the public record.

894 **Code availability**

895 The full pipeline code, preprocessing (reproducible from the public source
896 databases), patient-level split generation, model training, the calibration-
897 method implementations, and all statistical analyses, together with the
898 environment lock file, the patient-level split indices for all five dataset
899 variants and five random seeds, and selected per-experiment result artifacts,
900 is publicly available at [https://github.com/gt17641001169-design/](https://github.com/gt17641001169-design/ecg-calibration-transportability)
901 **ecg-calibration-transportability**. Trained model checkpoints for
902 the 60-experiment transfer grid are provided as release assets of the same
903 repository. The repository will be archived on Zenodo with a DOI upon
904 acceptance. No new raw waveforms were collected or generated; the
905 patient-level split indices and label-mapping tables released with the code
906 fully determine the data configuration used in every experiment.

907 **Declaration of generative AI and AI-assisted technologies in the** 908 **writing process**

909 All scientific work reported in this study—the study design, the pre-
910 registered protocol and its amendments, the implementation and execution
911 of the 60-experiment transfer grid, the statistical analysis, the interpretation
912 of the results, and the writing of the manuscript—was performed by the
913 author. During the final preparation of the manuscript, the author used a
914 large language model assistant (GLM) solely as a language-editing aid, to
915 improve wording and readability of text the author had written. It was not
916 used to generate scientific content, experimental data, analyses, figures, or

917 conclusions. After using this tool, the author reviewed and edited the text as
918 needed and takes full responsibility for the content of the published article.

919 **CRedit authorship contribution statement**

920 **Toni Guan:** Conceptualization, Methodology, Software, Validation, For-
921 mal analysis, Investigation, Data curation, Writing – original draft, Writing
922 – review & editing, Visualization, Project administration.

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930 **Declaration of competing interest**

931 The author declares no competing financial or personal interests.

932 **Supplementary materials.** The 60 per-seed reliability diagrams (6
933 transfer pairs $\times$ 2 architectures $\times$ 5 seeds; calibration curves, reliability dia-
934 grams, and Δ ECE bars per experiment) are provided as a separate supple-
935 mentary file (`supplementary.pdf`, §1) and are additionally released with the
936 code repository.

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Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC	Incep	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	ResNet	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap→PTB-XL	Incep	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓
	ResNet	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC→Chap	Incep	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	ResNet	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (Inception-Time). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat:* 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155